

PH 511: Mathematical Physics

Course Code	PH 511
Course Name	Mathematical Physics
L-T-P-C	3-0-0-3
Prerequisites	Undergraduate physics courses
Intended for	Core for M.Sc. and I-Ph.D. in Physics; Elective for B.Tech. 3rd and 4th Year.
Mutual Exclusion	NA
Approval	9th Senate (May 2016), in the Minor in Physics curriculum
Revised	64th BoA (May 2026) - credits changed to 3-0-0-3

Preamble

Mathematical physics provides a firm foundation in various mathematical methods developed and used for understanding different physical phenomena. This course provides mathematical tools to address formalisms used in the core course of master's level physics program.

Course Modules

- **1. Graphical Analysis and Vector Calculus Review:** Two-dimensional plots of simple functions, Review of Vector Calculus in Curvilinear coordinates. [4 lectures]
- **2. Linear Vector Spaces:** Linear vector spaces, Gram-Schmidt orthogonalization, Self-adjoint, Unitary and Hermitian Operators, Transformation of Operators: Unitary and Similarity Transformations, Eigenvalue Equation, Hermitian Matrix Diagonalization, Dirac Delta function, Infinite-dimensional Vector Spaces, Hilbert spaces. [11 lectures]
- **3. Ordinary Differential Equations:** Review of Second order Linear ODE, Series Solution: Frobenius Method, Inhomogeneous linear ODE, Sturm Liouville equation, Hermitian operators: Eigenvalue Problem. [8 lectures]
- **4. Special Functions:** Bessel, Neumann, Hermite, Legendre, Spherical Harmonics, Laguerre, Gamma and Beta functions. [5 lectures]
- **5. Complex Analysis:** Cauchy-Riemann conditions, Cauchy's Integral theorem, Laurent expansion, Singularities, Calculus of residues, evaluation of definite integrals. [9 lectures]
- **6. Fourier Analysis:** Fourier Series general properties and application, Integral transform, Properties of Fourier transform, Laplace transform, Convolution theorem. [5 lectures]

Laboratory/practical/tutorial Modules: No

Text Books

1. Arfken and Weber, *Mathematical methods for physicists* (Elsevier Academic Press, 7th edition, 2012)
2. Mary L Boas, *Mathematical Methods in Physical Sciences*, (Wiley 3rd edition, 2005)

References

1. V. Balakrishnan, *Mathematical Physics with Applications, Problems and Solutions* (Ane Books, 2023): Discussion of plotting of functions, Compact discussion of linear vector spaces
2. K. F. Riley and M. P. Hobson, *Mathematical Methods for Physics and Engineering: A Comprehensive Guide* (Cambridge India South Asian Edition, 2009)
3. J. Mathews and R. L. Walker, *Mathematical Methods for Physicists* (Imprint, New edition 1973)
4. F.W. Byron and R.W. Fuller, *Mathematics of Classical and Quantum Physics* (Dover Publication, New edition, 1992)
5. P. M. Morse and H. Freshbach, *Methods of theoretical Physics* Vol. I and II (McGraw-Hill, 1953)
6. E. Kreyszig, *Advanced Engineering Mathematics* (Wiley India Private Limited, 10th edition, 2003)
7. P. Dennery and A. Krzywicki, *Mathematics for Physicists* (Dover Publications Inc. 1996)
8. M. Spiegel, *Complex Analysis* (McGraw-Hill Education, 1980)
9. M. Spiegel, *Vector Analysis* (McGraw-Hill Education, 2nd Edition, 2016)